# Adverse Selection in the Annuity Market: A Behavioral Economics Perspective

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**Abstract**

This proposal aims to explore the phenomenon of adverse selection in annuity markets by explaining and incorporating the possible over-consumption behavioral patterns of individuals in their early working years. By classifying annuitants into two types: high tendency to over-consume (type 1: $t^1$) and low tendency to over-consume (type 2: $t^2$) in their early working ages, the ideas of how buyers and sellers interact in the annuity market at a certain time period can be illustrated by a 4-stage life cycle model from the perspective of Behavioral Economics.

**Keywords:** annuity market; adverse selection; behavioral economics; over-consumption

# 1 Introduction

Adverse selection has been a heated topic among scholars. In the context of the annuity market, adverse selection is usually caused by asymmetric information between annuitants and annuity providers. Annuitants are assumed to have private health information, and some annuitants with a higher likelihood to live longer will be more willing to buy annuities. As annuity providers lack access to annuitants' private health information, they can only impose a uniform pricing policy across all annuity products based on the average life expectancy of annuitants. To prevent a loss, annuity providers will charge a higher price for all annuitants, and some annuitants, especially those who have a shorter expected lifespan, will leave the annuity market (Walliser, J., 2000).

The above is the standard argument for how adverse selection exists in the annuity market. Although valid, it only considers asymmetric health information in the annuitant market. In this proposal, I would focus on the **existence of asymmetric information about the over-consumption tendency in the annuity market**, the relationship between (1) **life expectancy**, (2) **loss aversion**, (3) **present bias** and **tendency to over-consume in early working years**. The tendency to over-consume, just like health information, should be regarded as unobservable to the annuity providers. For simplicity, annuitants are classified into only two types, (1) annuitants with a high over-consumption tendency (type 1: $t^1$) and (2) annuitants with a low over-consumption tendency (type 2: $t^2$) in their early working years. I assumed they have exactly equal characteristics other than this. I will explain more on the design and testing of factors that correlates with over-consumption in next sections. At this moment, we just assume that $t^1$ **annuitants have a low expectation that they can survive in the later retirement stage, vice versa.** I further assume that only $t^1$ annuitants can have the update of their health signals after passing the early working years, but not $t^2$, since $t^2$ annuitants typically are more forward-looking and assumed to have an accurate estimate of their life expectancy at the beginning. In recent research, Lau & Zhang (2023) also incorporates heterogeneous early health signals to explain the adverse selection issue, which is a good starting point.

I will first focus on the scenario that $t^1$ annuitants find that they have

underestimated their life expectancy at the beginning. As they have been over-consuming, they are more inclined to buy a larger amount of annuity products in their later working years (next stage) than $t^2$ annuitants, in order to secure a stable income stream during their retirement periods. The detailed difference in purchasing decisions between them is as follows: As $t^2$ annuitants are accurately anticipating the life duration, they will start buying annuities since their early working years and remain buying the same amount of annuities in the later working years; On the contrary, $t^1$ annuitants are assumed not to buy any annuity in early working years because they are over-consuming, but they will buy a larger amount of annuity products compared to $t^2$ annuitants in later working years. It is important to relate this difference to adverse selection here. As discussed above, annuity providers rely on estimating the average survival probability of annuitants to determine the uniform price of annuity. If $t^1$ annuitants start buying annuities in later working years, the annuity providers have to re-estimate the average life expectancy of annuitants. To mitigate the risk of loss, annuity providers would probably set up a higher annuity price compared to the price determined in early working years of the same class of annuitants. It is expected that the combined effect of over-consumption and life expectancy underestimation in early working years **will amplify the effect of adverse selection in the market with a bigger rise in price** when solely comparing the effect of life expectancy underestimation. Then, in later working years, $t^2$ annuitants will probably reduce the amount of annuity purchased and even leave the annuity market.

# 2 Literature Review

## 2.1 Asymmetric Health Information and Adverse Selection in the Annuity Market

In the previous section, I have bridged the relationship between the over-consumption tendency of annuitants and other behavioral factors. It is crucial to review how scholars approach the adverse selection problem in the annuity market by some mathematical models with a traditional economic approach first.

Yarri's research on "Uncertain Lifetime, Life Insurance, and the Theory of

the Consumer" (1965) is considered to be one of the most classic and influential journal articles in the discussion of annuity market and annuitants' purchase decisions. One of the main assumptions of this paper is that he assumed the annuity market is perfect, and therefore no asymmetric information exists in the market. Also, the lifespan of annuitants are uncertain. He assumed that individuals always seek to [1] maximize the expected lifetime utility under wealth constraint: that is:

$$\max_{c} \; V(c) = \int_0^T \alpha(t) g[c(t)] \; dt$$

$$\text{subject to} \;\; S(T) \geq 0$$

where

$$S(t) = \int_0^t e^{\int_\tau^t j(x) \; dx} \{m(\tau) - c(\tau)\} \; d\tau$$

The above model is the general mathematical model the author used to explain how individuals make optimal consumption decisions over time. In the context of annuity purchase decisions, the consumers are buying actuarial notes, which is equivalent to buying annuities according to the author.

While Yarri's work has built a solid theoretical framework for the later research, his work may not be able to explain the adverse selection in the annuity market, as from the previous section we know that adverse selection is mainly caused by the asymmetric information in the annuity market. Agar Brugiavini's paper on "Uncertainty Resolution and the Timing of Annuity Purchases" (1993) also credited Yarri's work on the interest of the annuity market functioning. However, instead of setting continuous time period for the decision of annuity purchase, she only restricted the time period into

[1] The mathematics idea behind this problem is simple: Annuitants discounted future utility at time $t$ ($g[c(t)]$) by $\alpha(t)$. Assuming a continuous time interval, annuitants sums the expectation (by $\int$) from beginning ($t = 0$) to the end of life ($t = T$). Of course, annuitants will face certain wealth constraints (net assets $S(t)$), and it should be impossible for them to have negative wealth at the end of their life. Therefore, $S(T) \geq 0$ (or $= 0$) by the assumption of non-satiation. At time $\tau$, where $0 \leq \tau \leq t$, annuitants will put his/her saving (denoted by $m(\tau)$ rate of earnings at time $\tau$ minus consumption at $\tau$) into somewhere else (i.e., bank) to earn compounded interests ($j(\tau)$ is the rate of interest at time $\tau$). There are actually many more assumptions made, for example, the annuitants' preference are independent over time; c is bounded and measurable, $c(t) \geq 0 \; \forall t \in [0, T]$, and $S(t) = \int_0^T e^{\int_t^T j(x) \; dx} \{m(t) - c(t)\} \; dt = 0$

three discrete stages — youth (period 0), working years (period 1) and retirement (period 2). The annuitants only faced lifespan uncertainty in period 2. The major advantage of categorizing an annuitants' life in this way is that we can focus on the purchasing behavior of annuitants in a more systematic way. This research will adopt Agar's work (1993) as a theoretical foundation while the model will be extended into 4-stage model (which is to be discussed later). For the benefit of understanding my proposed model, the basic details of Agar's model should be discussed here first. The authors first assumed annuitants' likelihood to survive in period 2 follows a random probability distribution from $p_L$ to $p_H$. The utility maximization problem at different periods is solved by the method of **backward induction**:

In **period 2**, as the author assumed the annuitants have [2]no bequest motives (a standard assumption), annuitants will exhaust all the savings as it is the last stage of their life. Therefore, the timings that annuitants really need to make purchase decisions are period 1 and period 0.

In **period 1**, The annuitant solves the following utility maximization problem:

$$\max U(C_{1i}) + \beta p_i U(C_{2i})$$

$$\text{s.t.} \quad Q_i C_{1i} + C_{2i} = Q_i(Ry_0 + y_1) + (R^* - RQ_i)A_0, \quad i = H, L.$$

In **period 0**, The annuitant solves the following utility maximization problem:

$$\max J(A_0, \alpha) = \alpha\{U(C^{1H}) + \beta p_H U(C^{2H})\} + (1-\alpha)\{U(C^{1L}) + \beta p_L U(C^{2L})\}$$

## Over-consumption and Behavioral Economics

The above mathematical formulations are from the traditional economics perspective. However, numerous empirical evidence have shown that humans are not always behaving rationally. Then, behavioral economics comes into discussions. Jeffrey R. Brown (2007) suggested that psychological biases may be able to explain why consumers are not willing to buy annuities despite

[2]In fact, some researchers are interested in how the existence of bequest motives can affect the annuitant's decisions. In "Bequest Motives and the Annuity Puzzle" (2011), the author Lee M. Lockwood shows that if people have plausible bequest motives, it is beneficial for them not to buy any annuity.

annuities can be welfare-enhancing under certain conditions. Mitchell et. al. (2003) argues that the major behavioral factor that explains the low demand for annuity is **loss aversion**. The rationale is that consumers worry they may not be able to live long, and because it is impossible to transfer the annuity to the heir after the death of annuitants, this will incur a loss (for the heir). Besides loss aversion, the authors also mention other type of behavioral elements in the discussion. The author argues that retirees may seriously discount the future continuous benefits brought by buying life annuity, and they are inclined to using the lump sum accruals instead. Although the authors does not specifically mention the type of behavioral biases here, from the behavioral economic theory, this belongs to the **present bias** problem.

# 3 Hypotheses

I would like to extend the present bias issue to the stage of consumers in their early working years. At early working years, individuals may have poor planning in the future, and perhaps they do not want to form a new family. Therefore, **they have higher tendency to over-consume** in that period. However, if they have a well planning about future, their tendency to over-consume is low. This fulfills my initial classifications of annuitants into $t^1$ annuitants (annuitants with high tendency to over-consume) and $t^2$ annuitants (annuitants with low tendency to over-consume). It is natural to purpose the hypothesis that the tendency of annuitants to over-consume is negatively correlated to their life expectancy. The rationale is as follows: If people do not expect they can live long, they will tend to spend more and save less during their young ages (i.e., early working years), vice versa. Also, the tendency of annuitants to over-consume should be positively correlated to the level of loss aversion and present bias. To substantiate this hypothesis, experimental design and empirical testing should be performed in the research process. The causal relationship between these two variables should also be investigated.

# 4 Methodology

To proceed, we need to start from a existing theoretical framework. As mentioned, this research will adopt Agar's model as a theoretical framework. The major modifications will be using a hyperbolic discounting factor instead of using traditional exponential discounting factors. Also, the life cycle of an annuitants will be 4-staged instead of 3-staged, and the stages are early working years (stage 0), later working years (stage 1), early retirement years (stage 2) and later retirement years (stage 3). It is also important to measure how life expectancy, loss aversion and present bias correlated with the tendency to over-consume. The correlation can be investigated by designing some surveys for the consumers/annuitants/potential annuitants. In addition, Instrumental variable regressions (i.e., 2-stage least square regressions) can be further built to explore the possibility of causal relationships between each pair of variables.

# 5 Expected Contributions

This research paper is expected to explain detailedly how behavioral biases can lead to the existence of over-consumption of annuitants at early working years, and how over-consumption issue can exacerbate the problem of adverse selection in the annuity market. As the majority of papers that discusses adverse selection in the annuity market are from the traditional economic view, this research should be able to provide new and meaningful insights to this problem.

# 6 References

Agar Brugiavini, Uncertainty resolution and the timing of annuity purchases, Journal of Public Economics, Volume 50, Issue 1, 1993, Pages 31-62, ISSN 0047-2727, https://doi.org/10.1016/0047-2727(93)90059-3.

Brown, J. R. (2007). *Rational and Behavioral Perspectives on the Role of Annuities in Retirement Planning* : https://doi.org/10.3386/w13537

Lau, S.-H. P. and Zhang, Q. (2023), Heterogeneous early health signals and deferred annuities, Working Paper

Mitchell, Olivia S. and Utkus, Stephen P., Lessons from Behavioral Finance for Retirement Plan Design (October 2003). PRC Working Paper No. 2003-6, Available at SSRN: https://ssrn.com/abstract=464640 or http://dx.doi.org/10.2139/ssrn.464640

Walliser, J. (2000). Adverse Selection in the Annuities Market and the Impact of Privatizing Social Security. *The Scandinavian Journal of Economics*, *102*(3), 373–393. http://www.jstor.org/stable/3440626

Yaari, M. E. (1965). Uncertain Lifetime, Life Insurance, and the Theory of the Consumer. *The Review of Economic Studies*, *32*(2), 137–150. https://doi.org/10.2307/2296058